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Arithmetic

## ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int a=sc.nextInt();
    sc.nextLine();
    int b=sc.nextInt();
    int add=a+b;
    int sub=a-b;
    int mul=a*b;
    int div=a/b;

    System.out.println("Addition: "+add);
    System.out.println("Subtraction: "+sub);
    System.out.println("Multiplication: "+mul);
    System.out.println("Division: "+div);
  }
}
```

OUTPUT

```
Addition: 24
Subtraction: 16
Multiplication: 80
Division: 5
```

20
4

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QUESTIONS 31 / 31 completed

MAX

## MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java

Run Save

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int a=sc.nextInt();
    sc.nextLine();
    int b=sc.nextInt();
    sc.nextLine();
    int c=sc.nextInt();

    if(a>b && a>c){
      System.out.println("A is Greater");
    }
    else if(b>a && b>c){
      System.out.println("B is Greater");
    }
    else{
      System.out.println("C is Greater");
    }
  }
}
```

OUTPUT

```
C is Greater
```

INPUT

```
5
9
45
```

PROGRAM EDITOR

Java Run Save

INPUT

Leap

## LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int Leapyear=sc.nextInt();
    if((Leapyear%400==0) || (Leapyear%4==0 && Leapyear%100!=0)){
      System.out.println("Leap Year");
    }
    else{
      System.out.println("Not a Leap Year");
    }
  }
}
```

2023

OUTPUT

```
Not a Leap Year
```

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Fibonacci

## FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int a=0,b=1;

    for(int i=1;i<=n;i++){
      System.out.print(a + " ");
      int c=a+b;
      a=b;
      b=c;
    }
  }
}
```

OUTPUT

```
0 1 1 2 3 5 8 13 21 34
```

INPUT

10

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PROGRAM EDITOR

Java

Run Save

INPUT

Factorial

## FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    long fact=1;

    for(int i=1;i<=n;i++){
      fact=fact*i;
    }
    System.out.println(fact);
  }
}
```

5

OUTPUT

```
120
```

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RT-Pattern

## RT-PATTERN

Write a Java program to print the following pattern using loops:*
**
***
****
*****

```java
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();

    for(int i=1;i<=n;i++){
      for(int j=1;j<=i;j++){
        System.out.print("* ");
      }
      System.out.println();
    }
  }
}
```

3

OUTPUT

```
*
* *
* * *
```

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Search

## SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int[] arr={10,20,30,40,50};
    System.out.println("Enter the Element to Search:");
    int key=sc.nextInt();

    int low=0;
    int high=arr.length-1;
    int found=-1;

    while(low<=high){
      int mid=(low+high)/2;
      if(arr[mid]==key){
        found=mid;
        break;
      }
      else if(arr[mid]<key){
        low=mid+1;
      }
```

40

OUTPUT

```
Enter the Element to Search:
Element Found at index:3
```

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GCD

## GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    System.out.println("Enter First Number:");
    int a=sc.nextInt();
    System.out.println("Enter Second Number");
    int b=sc.nextInt();

    while(b!=0){
      int temp=b;
      b=a%b;
      a=temp;
    }
    System.out.println("GCD:"+a);
  }
}
```

OUTPUT

```
Enter First Number:
Enter Second Number
GCD:10
```

20
10

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Prime

## PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    boolean isprime=true;

    if(n<=1){
      isprime=false;
    }
    for(int i=2;i<=Math.sqrt(n);i++){
      if(n%i==0){
        isprime=false;
        break;
      }
    }
    if(isprime){
      System.out.println("Prime Number");
    }
    else{
      System.out.println("Not Prime");
```

INPUT

2

OUTPUT

```
Prime Number
```

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Sort

## SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int[] arr={54,12,65,61,43};
    int n=arr.length;

    for(int i=0;i<n-1;i++){
      int min=i;
      for(int j=i+1;j<n;j++){
        if(arr[j]<arr[min]){
          min=j;
        }
      }
      int temp=arr[min];
      arr[min]=arr[i];
      arr[i]=temp;
    }
    System.out.println("Sorted Array:");
    for(int i=0;i<n;i++){
      System.out.print(arr[i]+" ");
```

Your INPUT goes here!

OUTPUT

```
Sorted Array:
12 43 54 61 65
```

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Table

## TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run Save

```java
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();

    for(int i=1;i<=10;i++){
      System.out.println(n+"*"+i+"="+(n*i));
    }
  }
}
```

OUTPUT

INPUT

10

OUTPUT

```
10*1 10
10*2 20
10*3 30
10*4 40
10*5 50
10*6 60
10*7 70
10*8 80
10*9 90
10*10 100
```

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Factorial

## FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

```
import java.util.Scanner;
public class Main{
  static long factorial(int n){
    if(n==0 ||n==1){
      return 1;
    }
    else{
      return n*factorial(n-1);
    }
  }
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    System.out.println("Factorial="+factorial(n));
  }
}
```

OUTPUT

```
Factorial=120
```

5

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LT-Pattern

## LT-PATTERN

Write a Java program to print the following pattern using loops:*****
****
***
**
*

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();

    for(int i=n;i>=1;i--){
      for(int j=1;j<=i;j++){
        System.out.print("*");
      }
      System.out.println();
    }
  }
}
```

4

OUTPUT

```
****
***
**
*
```

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LCM

## LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);

    int a=sc.nextInt();
    int b=sc.nextInt();

    int x=a;
    int y=b;

    while(y!=0){
      int temp=y;
      y=x%y;
      x=temp;
    }
    int gcd=x;
    int lcm=(a*b)/gcd;
    System.out.println(lcm);
  }
}
```

10
12

OUTPUT

```
60
```

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D to B

## D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int binary[]=new int[32];
    int i=0;

    while(n>0){
      binary[i]=n%2;
      n=n/2;
      i++;
    }
    System.out.println("Binary:");
    for(int j=i-1;j>=0;j--){
      System.out.print(binary[j]);
    }
  }
}
```

13

OUTPUT

```
Binary:
1101
```

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B to D

## B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int binary=sc.nextInt();
    int decimal=0;
    int base=1;

    while(binary>0){
      int lastDigit=binary%10;
      decimal=decimal+lastDigit*base;
      base=base*2;
      binary=binary/10;
    }
    System.out.println("Decimal:"+decimal);
  }
}
```

1101

OUTPUT

```
Decimal:18
```

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Sum

## SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int sum=0;

    while(n>0){
      int digit=n%10;
      sum=sum+digit;
      n=n/10;
    }
    System.out.println(sum);
  }
}
```

153

OUTPUT

```
9
```

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Product

## PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

```java
import java.util.Scanner;
public class Main{
  public static void main(String[] main){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int sum=1;

    while(n>0){
      int digit=n%10;
      sum=sum*digit;
      n=n/10;
    }
    System.out.println(sum);
  }
}
```

35

OUTPUT

```
15
```

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Power

## POWER

Write a Java program to find the power of a number using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int base=sc.nextInt();
    int exp=sc.nextInt();
    int pow=1;

    for(int i=1;i<=exp;i++){
      pow=pow*base;
    }
    System.out.println(pow);
  }
}
```

2
5

OUTPUT

```
32
```

Area Circle

## AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    double radius=sc.nextDouble();

    double area=3.14*radius*radius;

    System.out.println(area);
  }
}
```

7

OUTPUT

```
153.86
```

Shpere

## SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    double radius=sc.nextDouble();

    double volume=(4.0/3.0)*3.14*radius*radius*radius;

    System.out.println(volume);
  }
}
```

OUTPUT

```
1077.02
```

7

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Palindrome

## PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    String str=sc.nextLine();
    String rev="";

    for(int i=str.length()-1;i>=0;i--){
      rev=rev+str.charAt(i);
    }
    System.out.println(rev);
  }
}
```

monish

OUTPUT

```
hsinom
```

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Armstrong

## ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int original=n;
    int sum=0;

    while(n>0){
      int digit=n%10;
      sum+=digit*digit*digit;
      n=n/10;
    }
    if(sum==original){
      System.out.println("ArmStrong Number");
    }
    else{
      System.out.println("Not an ArmStrong Number");
    }
  }
}
```

**OUTPUT**

```
ArmStrong Number
```

153

RT-num.Pattern

## RT-NUM.PATTERN

Write a Java program to print the following pattern using loops:1
22
333
4444
55555

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();

    for(int i=1;i<=n;i++){
      for(int j=1;j<=i;j++){
        System.out.print(i+" ");
      }
      System.out.println();
    }
  }
}
```

3

OUTPUT

```
1
2 2
3 3 3
```

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Perfect

## PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
import java.util.Scanner;
public class Main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int sum=0;

    for(int i=1;i<n;i++){
      if(n%i==0){
        sum=sum+i;
      }
    }
    if(n==sum){
      System.out.println("Perfect Number");
    }
    else{
      System.out.println("Not a Perfect Number");
    }
  }
}
```

OUTPUT

```
Perfect Number
```

INPUT

28

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Max Array

## MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

```
public class Main{
  public static void main(String[] args){
    int[] arr={12,23,86,65,43};

    int max=arr[0];

    for(int i=1;i<arr.length;i++){
      if(arr[i]>max){
        max=arr[i];
      }
    }
    System.out.println(max);
  }
}
```

Your INPUT goes here!

OUTPUT

```
86
```

2nd Max

## 2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

```
public class Main{
  public static void main(String[] args){
    int[] arr={13,34,78,23,7};

    int largest=Integer.MIN_VALUE;
    int second=Integer.MIN_VALUE;

    for(int i=0;i<arr.length;i++){
      if(arr[i]>largest){
        second=largest;
        largest=arr[i];
      }
      else if(arr[i]>second && arr[i]!=largest){
        second=arr[i];
      }
    }
    System.out.println(second);
  }
}
```

Your INPUT goes here!

OUTPUT

```
34
```

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Min

## MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

```
public class Main{
  public static void main(String[] args){
    int[] arr={78,34,56,3,10};
    int min=arr[0];

    for(int i=0;i<arr.length;i++){
      if(arr[i]<min){
        min=arr[i];
      }
    }
    System.out.println(min);
  }
}
```

Your INPUT goes here!

OUTPUT

```
3
```

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2nd Min

## 2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

```
public class Main{
  public static void main(String[] args){
    int[] arr={13,34,78,23,7};

    int smallest=Integer.MAX_VALUE;
    int second=Integer.MAX_VALUE;

    for(int i=0;i<arr.length;i++){
      if(arr[i]<smallest){
        second=smallest;
        smallest=arr[i];
      }
      else if(arr[i]<second && arr[i]!=smallest){
        second=arr[i];
      }
    }
    System.out.println(second);
  }
}
```

Your INPUT goes here!

OUTPUT

```
13
```

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PROGRAM EDITOR

Java

Run

Save

INPUT

Sort-B

## SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

```java
public class Main{
  public static void main(String[] args){
    int[] arr={76,24,51,30,23};

    for(int i=0;i<arr.length-1;i++){
      for(int j=0;j<arr.length-i-1;j++){
        if(arr[j]>arr[j+1]){
          int temp=arr[j];
          arr[j]=arr[j+1];
          arr[j+1]=temp;
        }
      }
    }
    for(int num:arr){
      System.out.print(num+" ");
    }
  }
}
```

Your INPUT goes here!

OUTPUT

```
23 24 30 51 76
```

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Sort-I

## SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
public class Main{
  public static void main(String[] args){
    int[] arr={12,76,98,55,43};

    for(int i=1;i<arr.length;i++){
      int key=arr[i];
      int j=i-1;
      while(j>=0 && arr[j]>key){
        arr[j+1]=arr[j];
        j--;
      }
      arr[j+1]=key;
    }
    for(int i=0;i<arr.length;i++){
      System.out.print(arr[i]+" ");
    }
  }
}
```

OUTPUT

```
12 43 55 76 98
```

INPUT

Your INPUT goes here!